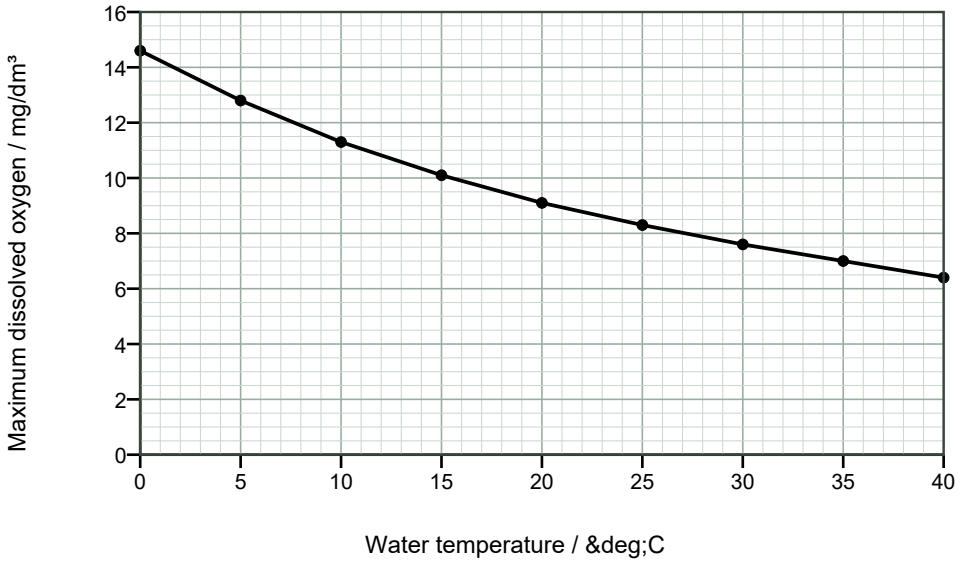
Questions by topic: Thermal & oil pollution - Set 1 (55 marks)

Answer all questions in the spaces provided.

0 1 Figure 1 shows the maximum concentration of dissolved oxygen in fresh water at different temperatures. A power station discharges warm cooling water into a river. The lowest dissolved oxygen concentration that trout can tolerate is 6.5 mg dm^{-3}.

Figure 1



0 1 . 1 Describe the relationship shown in Figure 1.

[1 mark]

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0 1 . 2 Use Figure 1 to determine the highest water temperature at which trout could survive if the water were fully saturated with oxygen.

[1 mark]

Answer ☐ °C

0 1 . 3 In summer the river upstream of the power station is at 20 °C and the effluent raises the temperature by 7 °C. Use Figure 1 to calculate the reduction in maximum dissolved oxygen concentration caused by the effluent.

[2 marks]

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Answer [ ] mg/dm³

0 1 . 4 Explain why the actual dissolved oxygen concentration downstream may be much lower than the values in Figure 1.

[2 marks]

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0 1 . 5 Describe **three** effects of warmer water on aquatic organisms other than reduced dissolved oxygen.

[3 marks]

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0 1 . 6 Describe how a cooling tower reduces the temperature of power station effluent water.

[1 mark]

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0 2 . 1 Give **three** sources of oil pollution of the sea other than tanker accidents.

[3 marks]

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0 2 . 2 Explain how an oil spill at sea can kill seabirds.

[3 marks]

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0 2 . 3 Explain **two** ways in which an oil slick can harm organisms that live in the water beneath it.

[2 marks]

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0 2 . 4 Explain why oil pollution is most serious when it occurs inshore or in an enclosed body of water.

[2 marks]

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0 3 Table 1 shows features of oil tanker design and operation that reduce the risk of oil pollution.

0 3 . 1 Complete Table 1 by explaining how each feature reduces oil pollution.

[5 marks]

Table 1

Feature	How it reduces oil pollution	
Double hull		
Inert gas system		
Separate oil and ballast tanks		
Twin engines / rudders		
Recirculation of oil during unloading (crude oil washing)		

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0 3 . 2 Explain how each of the following reduces the risk of tanker accidents.

[2 marks]

Offshore shipping routes .....

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GPS / Automatic Identification System .....

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0 3 . 3 Describe how a **bund wall** prevents oil pollution from a storage tank.

[2 marks]

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0 3 . 4 Explain the purpose of an oil interceptor on the drainage system of a car park.

[1 mark]

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0 4 In 2020 the tanker MV Wakashio ran aground on a coral reef off Mauritius and leaked about 1000 tonnes of fuel oil into a shallow lagoon near mangroves and a marine park.

0 4 . 1 Explain why inflatable booms were deployed immediately, and give **one** limitation of booms.

[3 marks]

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0 4 . 2 Describe how skimmers and absorbent materials remove oil, and state how the collected oil is disposed of.

[3 marks]

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0 4 . 3 Explain why dispersants were **not** used in the lagoon.

[2 marks]

0 4 . 4 Explain how bioremediation could help remove the oil that remained on the mangrove sediments, and suggest how it could be accelerated.

[2 marks]

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0 5 Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

Figure 1

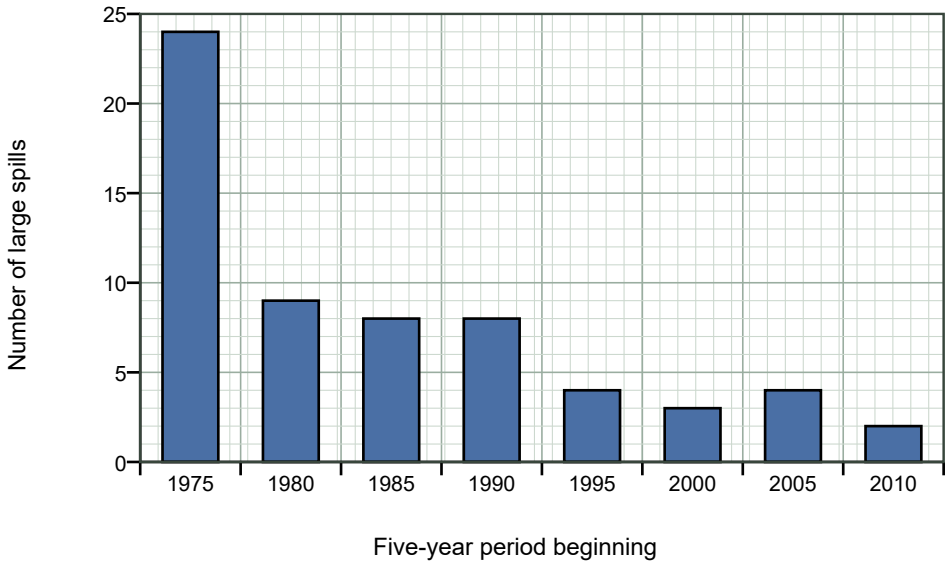


Table 1

Year	1975	1985	1995	2005	2015
Oil transported by sea / million tonnes	1 400	1 200	1 800	2 400	2 900

0 5 . 1 Calculate the percentage increase in oil transported by sea between 1975 and 2015.

[2 marks]

Answer ____ %

0 5 . 2 Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

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0 5 . 3 Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

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0 5 . 4 Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

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0 6 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 6 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 6 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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5

END OF QUESTIONS

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